Laporan Praktikum PBO: Getter, Setter, dan Encapsulation

Mata Kuliah: Pemrograman Berorientasi Objek / Kelas A
Nama Mahasiswa: Nabil Aulia Dika (Dezz)
NPM: 250810701100090
Topik: Pertemuan 4 - Getter and Setter

1. Tujuan Pembelajaran & Teori Dasar

Tujuan praktikum ini adalah memahami konsep *encapsulation* pada OOP, penggunaan access modifier private dan public, pembuatan method getter dan setter, serta implementasi validasi data pada setter agar integritas nilai data objek terlindungi dari modifikasi langsung di luar class.

2. Kode Program: Student.java

```
public class Student {
    private String nim;
    private String name;
    private String studyProgram;
    private double gpa;

    public Student(String nim, String name, String studyProgram) {
        this.nim = nim;
        this.name = name;
        this.studyProgram = studyProgram;
        this.gpa = 0.0;
    }

    public String getNim() { return nim; }
    public void setNim(String nim) { this.nim = nim; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public String getStudyProgram() { return studyProgram; }
    public void setStudyProgram(String studyProgram) { this.studyProgram = studyProgram; }

    public double getGpa() { return gpa; }

    public void setGpa(double gpa) {
        if (gpa >= 0.0 && gpa <= 4.0) {
            this.gpa = gpa;
        } else {
            System.out.println("Invalid GPA!");
            System.out.println("GPA must be between 0.0 and 4.0.");
        }
    }
}
```

3. Kode Program: Main.java

```
public class Main {
    public static void main(String[] args) {
        Student s1 = new Student("2024001", "Andi", "Informatika");
        System.out.println("NIM: " + s1.getNim() + ", Nama: " + s1.getName() + ", IPK: " + s1.getGpa());

        System.out.println("\nPengujian setGpa(3.75) [Valid]:");
        s1.setGpa(3.75);
        System.out.println("IPK baru: " + s1.getGpa());

        System.out.println("\nPengujian setGpa(5.0) [Invalid]:");
        s1.setGpa(5.0);
        System.out.println("IPK tetap: " + s1.getGpa());
    }
```

```
}
```

4. Output Eksekusi Terminal

```
================================================
   LATIHAN PEMROGRAMAN JAVA: GETTER & SETTER
================================================

[1] Membuat Objek Mahasiswa 1 (Andi):
Data Awal Mahasiswa 1:
NIM           : 2024001
Nama          : Andi
Program Studi : Informatika
IPK Awal      : 0.0

[2] Menguji setGpa(3.75) - IPK Valid:
IPK setelah diubah: 3.75

[3] Menguji setGpa(5.0) - IPK Tidak Valid:
Invalid GPA!
GPA must be between 0.0 and 4.0.
IPK sekarang (tetap nilai sebelumnya): 3.75
```